

CivicShield — Prototype Documentation & Tech Spec

AI-Enabled Verified Official Identity & Mobile Urban Sensing Platform • SIH 2026 PS 26124

Target Event: Smart India Hackathon 2026	Problem Statement: PS 26124
Backend API: Python FastAPI (Port 8000)	Frontend App: Next.js 14 App Router (Port 3000)
Local Execution: http://localhost:3000	Sensing Dashboard: http://localhost:3000/sih-sensing

1. System Capabilities & Overview

CivicShield is an enterprise-grade Smart Governance and Mobile Urban Intelligence platform created for **SIH 2026 Problem Statement PS 26124**. It converts public transport buses into continuous mobile sensing units detecting potholes, damaged signs, pedestrian dangers, and vehicle incidents while offering citizen QR official verification.

- **Mobile Fleet AI Sensing (/sih-sensing):** Features a 4-camera onboard video canvas switcher (Front, Rear, Side, Cabin), TensorRT YOLOv8 road defect overlays (Pothole 94.8%), ANPR license plate extraction, pedestrian danger alerts, and central Leaflet GIS hazard auto-pinning with automated GCC municipal repair tickets.
- **Official Identity Verification (/verify):** Browser QR scanner (html5-qrcode) verifying official badges against SQLite records, rendering ACTIVE/SUSPENDED status badges and anti-extortion warnings.
- **Rating Anti-Abuse Engine (/ratings):** Scikit-learn **IsolationForest ML model** (rating_anomaly.py) filtering multi-metric citizen feedback to detect rating manipulation bursts.
- **Multimodal Travel Planner (/mobility, /trip-manager):** Weighted utility route solver and **Google Gemini 1.5 Flash API** conversational trip manager.
- **Government Office Locator (/offices):** Python **Haversine distance calculator** sorting nearby Police, Hospital, EB, and Municipal offices.

2. Demo Role Accounts & Access Portals

Role	Demo Email	Password	Access Portals
Citizen	citizen@demo.com	Demo@123	/verify, /sih-sensing, /mobility, /offices
Official	officer@demo.com	Demo@123	/official/dashboard, /official/qr
Transport Authority	authority@demo.com	Demo@123	/authority/scheduling
Admin	admin@demo.com	Demo@123	/admin, /admin/analytics

3. AI / ML Implementation Matrix

Subsystem	Algorithm / Model	Input → Output Format	Implementation Status
Rating Anti-Abuse	IsolationForest	8 Metrics → Anomaly Score (-1 / 1)	IMPLEMENTED
Passenger Demand	RandomForestRegressor	Route, Hour, Weather → Count	IMPLEMENTED
O-D Transfer Pattern	DBSCAN Graph Clustering	Origin/Dest Coordinates → Cluster	IMPLEMENTED
Office Queue Time	Linear Regression	Visitors, Hour → Wait Mins	IMPLEMENTED
Conversational AI	Google Gemini 1.5 Flash	Text Query → Trip Plan	IMPLEMENTED
Mobile Fleet AI Sensing	YOLOv8s + TensorRT (Sim)	Video Frame → BBox, Conf, GPS	PROTOTYPE / DEMO

4. SIH Evaluator 11-Step Demonstration Story (BUS 104A)

- **Step 1:** Select BUS 104A (TN-01-N-9842) running on Route 70H.
- **Step 2:** Open Onboard 4-Camera Feeds (Front Road, Rear Traffic, Side Obstacle, Cabin Safety).
- **Step 3 & 4:** Computer vision detects severe asphalt pothole on Front camera (Class POTHOLE_SEVERE, 94.8% confidence).
- **Step 5:** Edge AI attaches GPS coordinates (13.0067°N, 80.2020°E) and microsecond ISO timestamp.
- **Step 6 & 7:** Hazard marker auto-pins onto central Leaflet GIS map and streams live to Command Dashboard.
- **Step 8:** Traffic Analytics displays vehicle counting (1,420 Bikes, 580 Autos, 1,120 Cars) and Kathipara delay (+14 mins).
- **Step 9:** Pedestrian Safety flags school children entering dangerous non-zebra zone near Koyambedu Metro.
- **Step 10:** ANPR Incident extracts license plate TN-09-AB-1234 driving at 74 km/h in 40 km/h zone.
- **Step 11:** Automated Municipal Work Order #GCC-ROAD-4092 dispatched to Greater Chennai Corporation.

5. Complete Technical Stack Summary

- **Frontend:** Next.js 14.2.35 (App Router), React 18.3, TypeScript 5.6, Tailwind CSS 3.4, Leaflet 1.9.4 GIS, HTML5-QRCode.
- **Backend:** Python 3.11/3.14, FastAPI 0.110, Uvicorn ASGI Server, SQLAlchemy 2.0 ORM, FastAPI WebSockets.
- **AI / ML:** Scikit-Learn (IsolationForest, RandomForest, DBSCAN), Google Gemini 1.5 Flash SDK, NVIDIA Jetson Edge AI Simulator.
- **Database:** Relational SQLite Database file (civicai.db) with 8 entities.